An Information-Theoretic View of Cache-Aided Networks: Part 1 – Coded Caching

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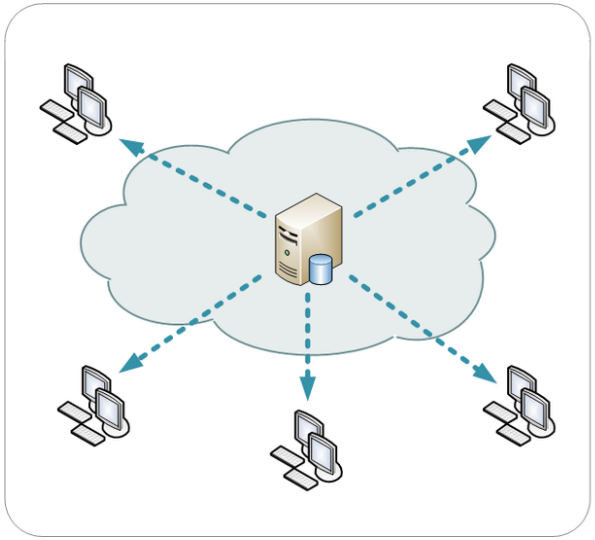
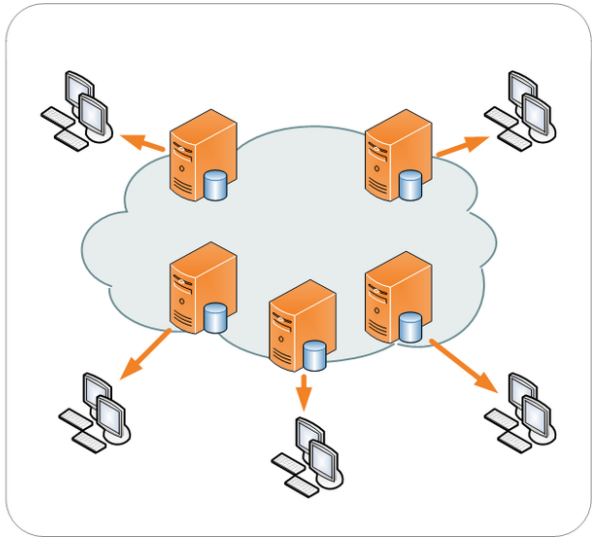
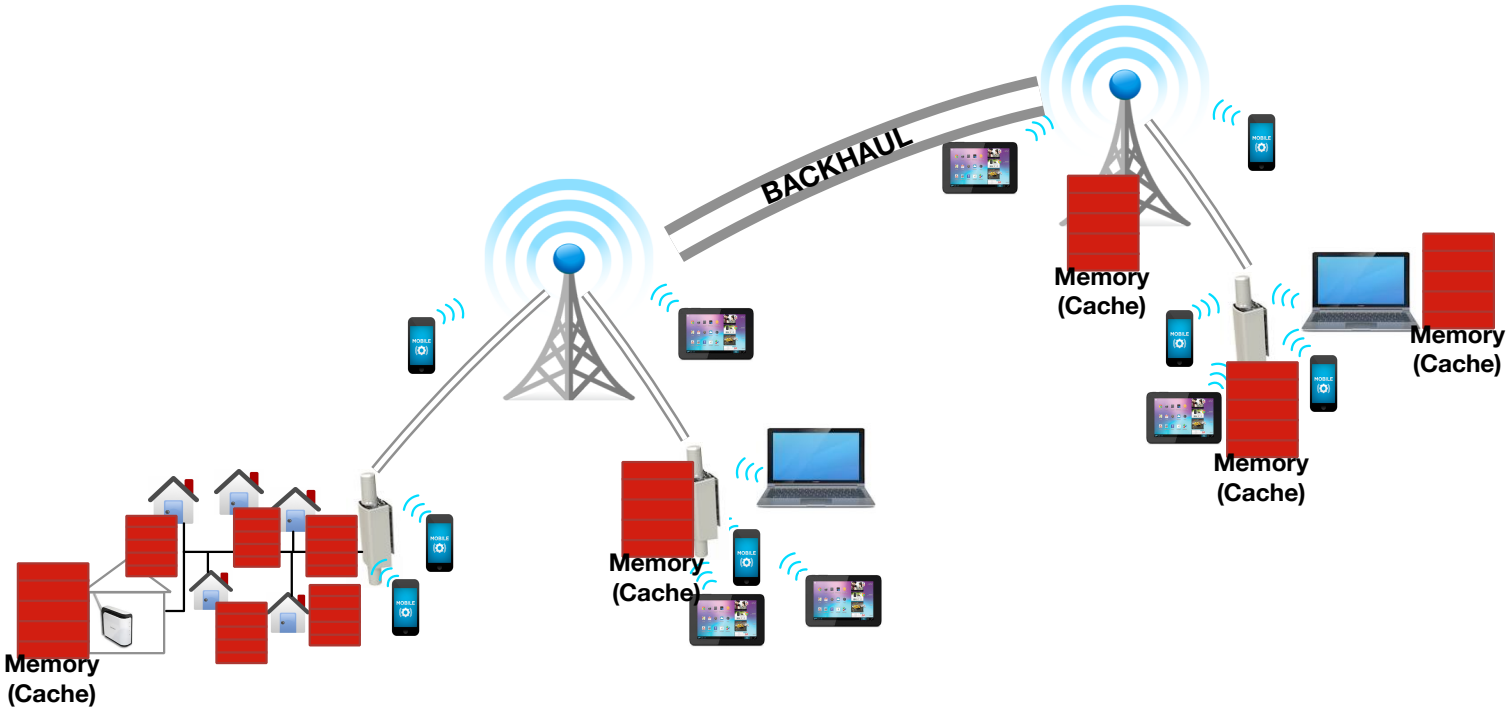
Content Delivery Networks





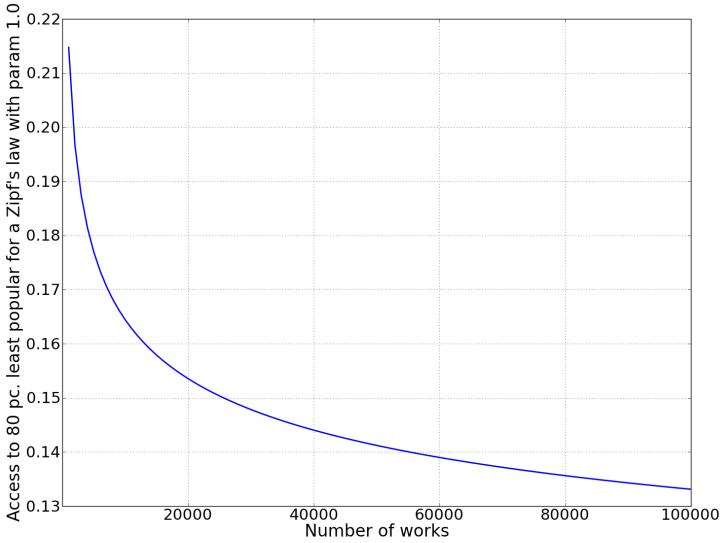
- Store contents in caches before file demands even known
- Reduce network load and latency during high-congestion periods
- Idea useful if certain files very popular and known in advance

Distributed Caches: Promising Solution for Cellular Networks



- Can cache at main BSs, picoBSs, femtoBSs, or directly at end users

File Popularities



- Static file popularity follows a Zipf distribution $P(x) = Cx^{-\alpha}$
- Evolution of file popularities (youtube videos) can also be predicted

Use pro-active caching to improve cellular systems!

- decrease network load
- decrease latency

Information-Theoretic View of Caching

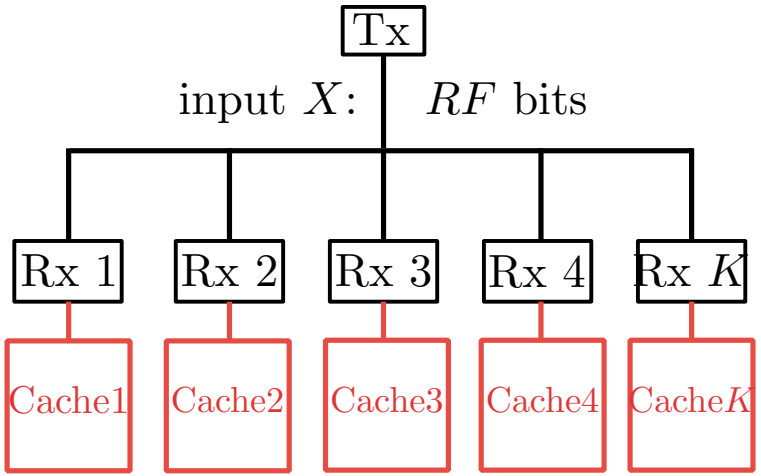
- Go beyond obvious *local* caching gain
- Create coding opportunities through smart cache placement
- Exploit multi-cast opportunities

  $\rightarrow$ serve many users/demands with same signals
- Global caching gain

  $\rightarrow$ receivers can profit from other receivers' cache memories

A Simple Network

<u>Library:</u> Files $W_1, W_2, \ldots, W_N$ of F bits each



- All files equally popular $\rightarrow$ we consider only most popular files
- All files equally large.
- Before the actual transmission there is an idle period where the transmitter (server) can fill the receivers' cache memories.
- Cache placement phase only constrained by memory sizes.

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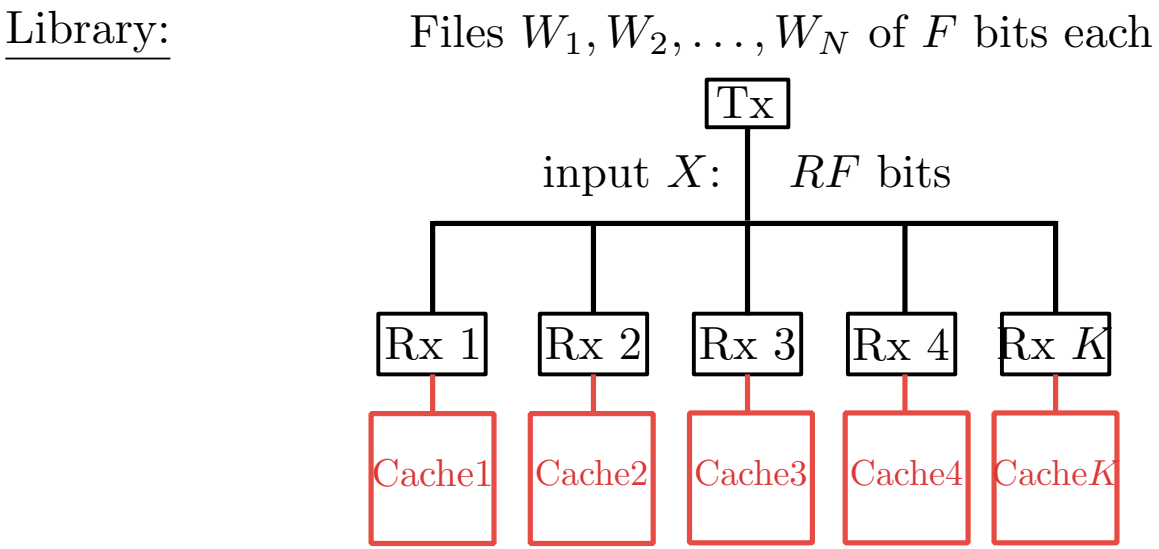
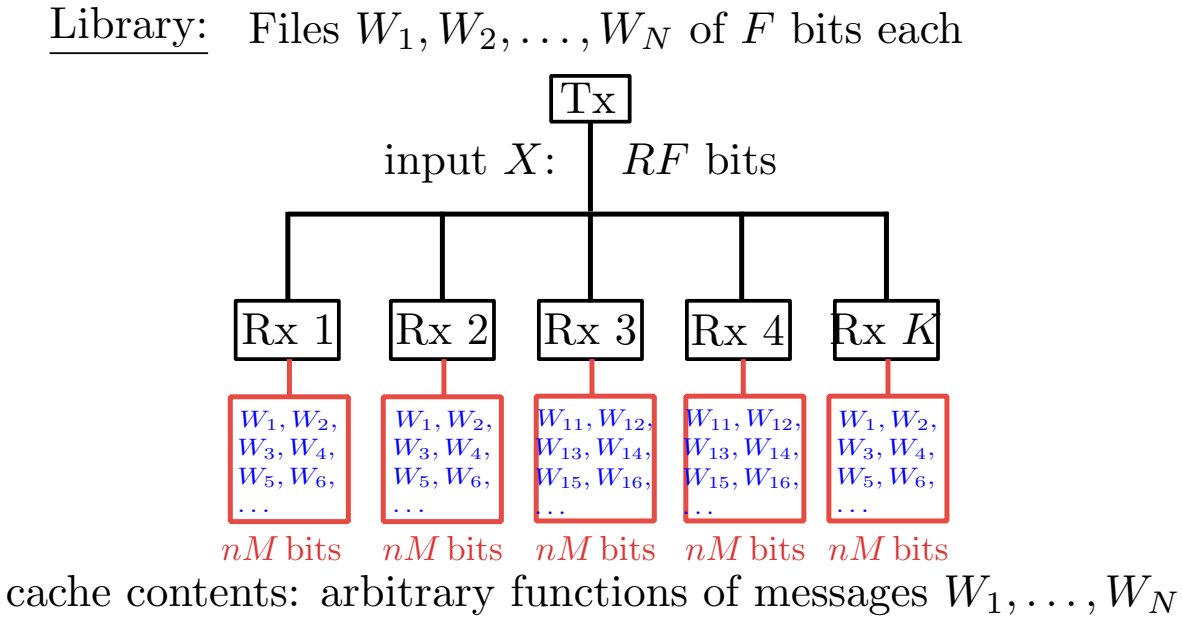
[1] M. A. Maddah-Ali, U. Niesen, "Fundamental Limits of Caching." *IEEE Transactions on Information Theory.*

A Simple Network



Communication in two phases:

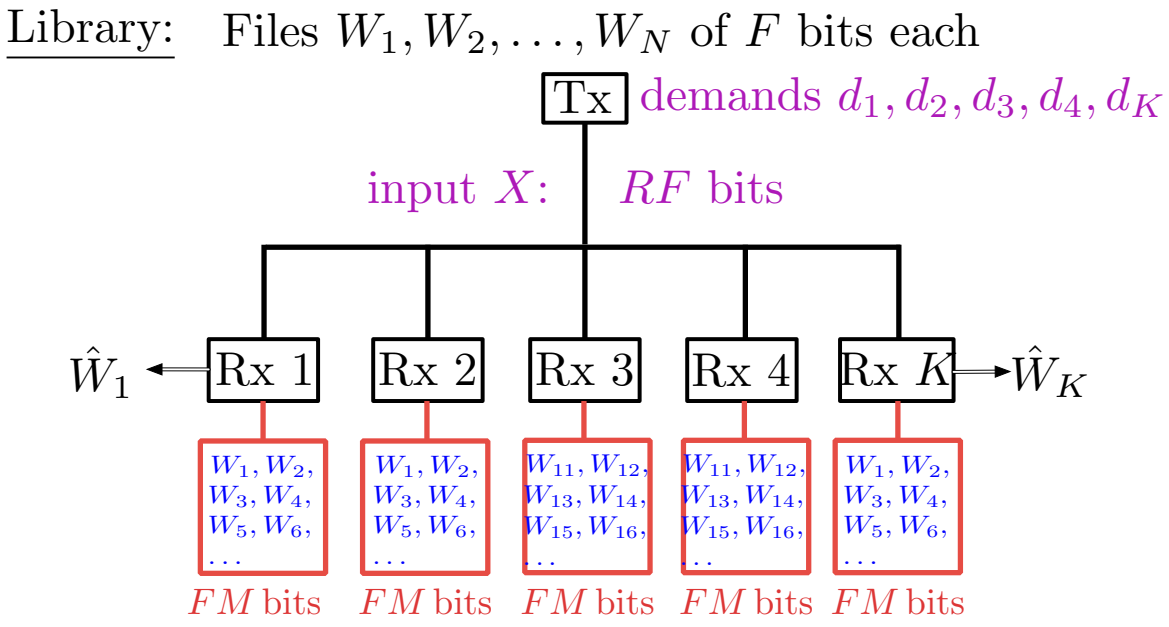
A Simple Network



Communication in two phases:

- Placement phase: Tx fills caches without knowing which receiver demands which message

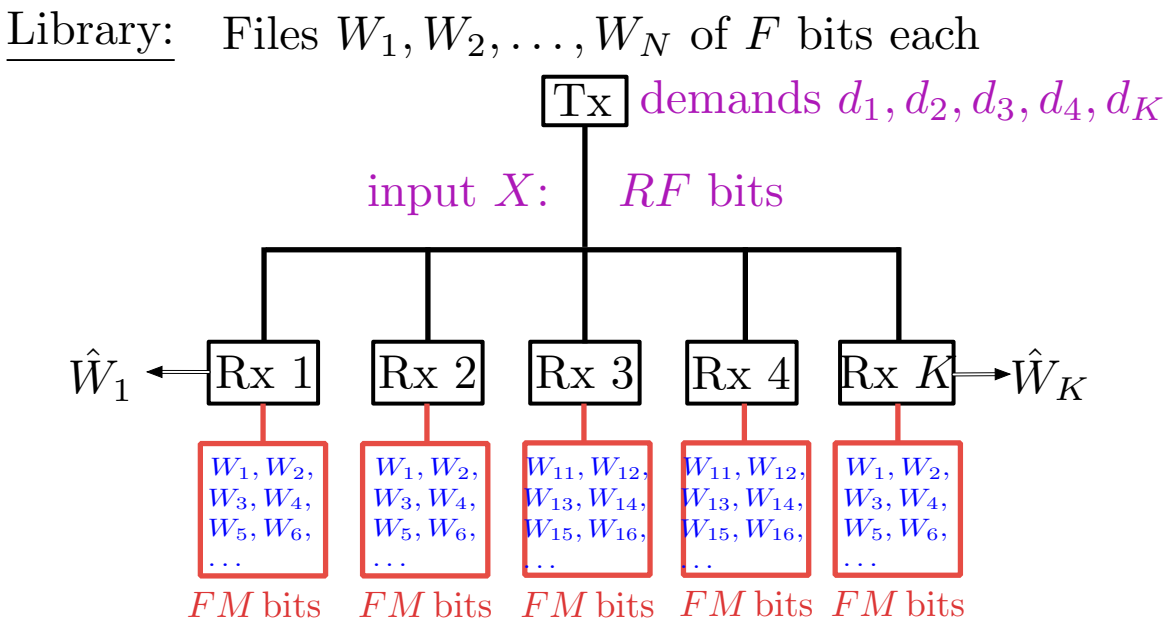
A Simple Network



Communication in two phases:

- Placement phase: Tx fills caches without knowing which receiver demands which message
- Delivery phase:
  - Receiver k wants file W_{d_k} $\rightarrow$ sends demand d_k to transmitter
  - Tx describes $W_{d_1}, \ldots, W_{d_K}$ to Rxs $1, \ldots, K$ through input X
  - Tx describes demands $d_1, \ldots, d_K$ to all receivers

Formal Problem Description



Fundamental Rate-Memory Tradeoff $R^\star(M)$

$$R^\star(M) \quad := \quad \min\big\{R\colon \text{ such that for } (R,M) \text{ goal can be achieved for } \textit{all demands } d_1,\ldots,d_K.\big\}$$

Some properties:

- $R^\star(M)$ is decreasing in M.
- $R^\star(M)$ is bounded above by $\min\{N,K\}$. Moreover:

$$R^\star(M=0) = \min\{N,K\}.$$

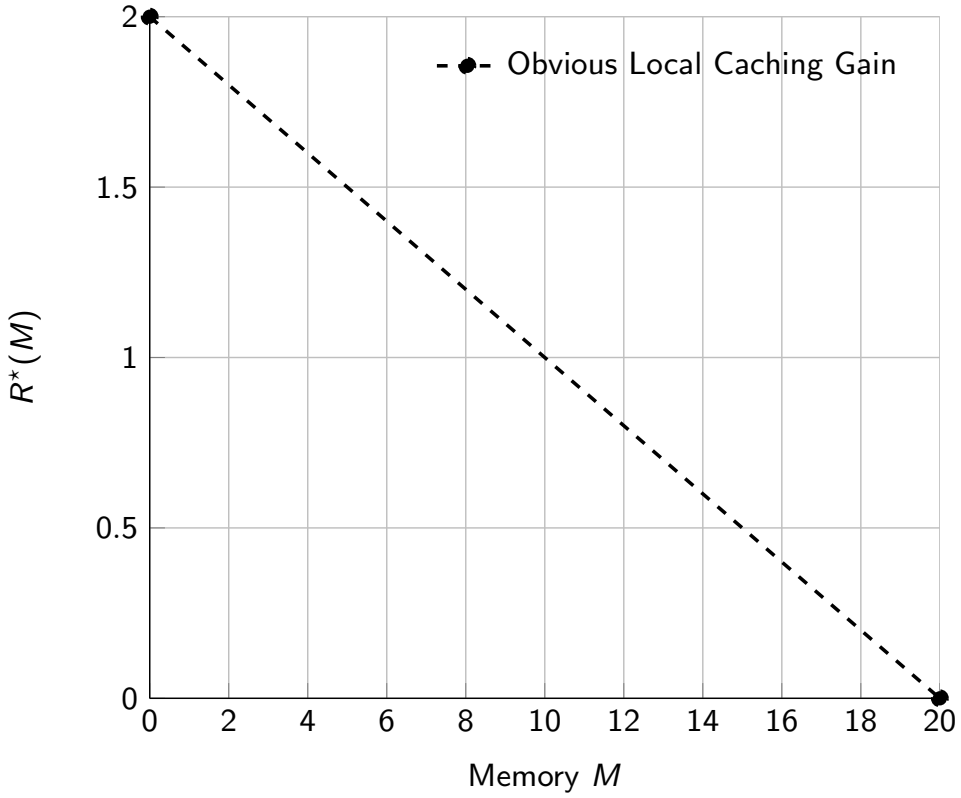
- $R^\star(M)$ is nonnegative. Moreover:

$$R^\star(M) = 0, \qquad \forall M \geq N.$$

Obvious Upper Bound on $R^\star(M)$

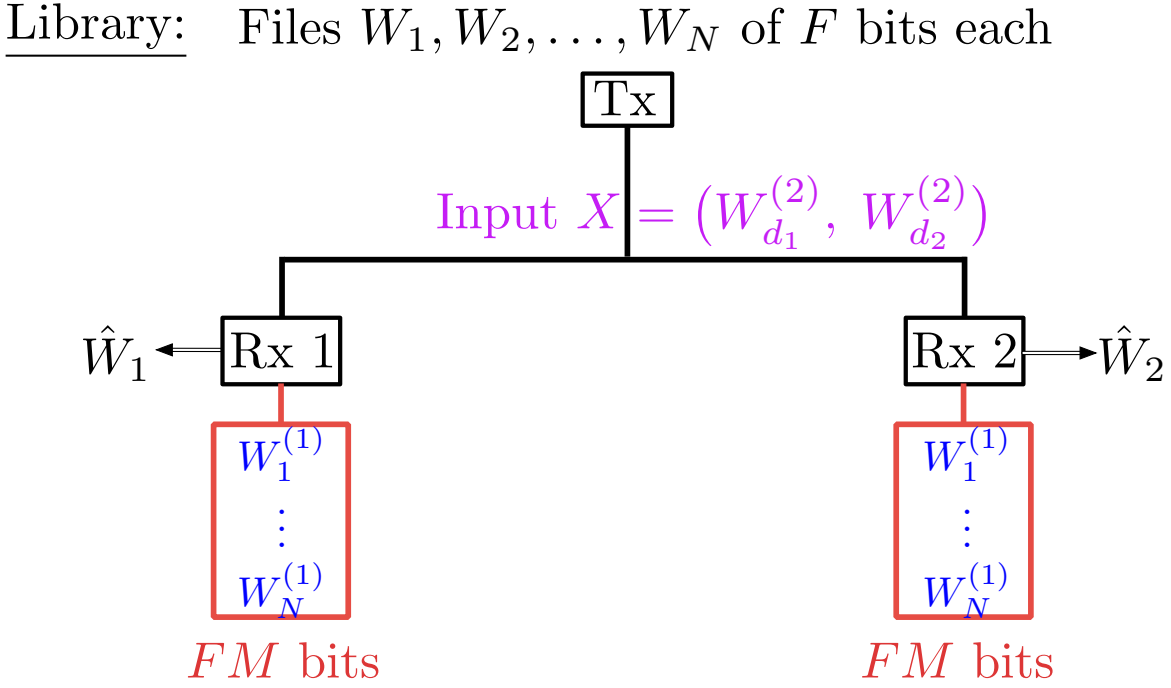
$$R^\star(M) \leq \min\{K, N\}\left(1 - \frac{M}{N}\right).$$

- $N = 20$ files and $K = 2$ Users



- Achieved through time/memory sharing or by the following naive scheme...

Naive Scheme for $K = 2$ Receivers



- Split W_d into two parts $(W_d^{(1)}, W_d^{(2)})$ of sizes $\frac{FM}{N}$ and $F\left(1 - \frac{M}{N}\right)$ bits
- For $d = 1, \ldots, N$: cache part $W_d^{(1)}$ at both Rxs 1 and 2
- Delivery input $X = (W_{d_1}^{(2)}, W_{d_2}^{(2)})$ (if $d_1 \neq d_2$)

In the worst-case, rate needs to be $R = 1$.

Naive Scheme for K Receivers

- Split W_d into two parts $(W_d^{(1)}, W_d^{(2)})$ of sizes $F\frac{M}{N}$ and $F\big(1-\frac{M}{N}\big)$ bits
- For $d = 1, \dots, N$: cache part $W_d^{(1)}$ *at all* rxs
- Deliver part $W_d^{(2)}$ for each demanded message W_d.
  - If $K \geq N$, in the worst case:

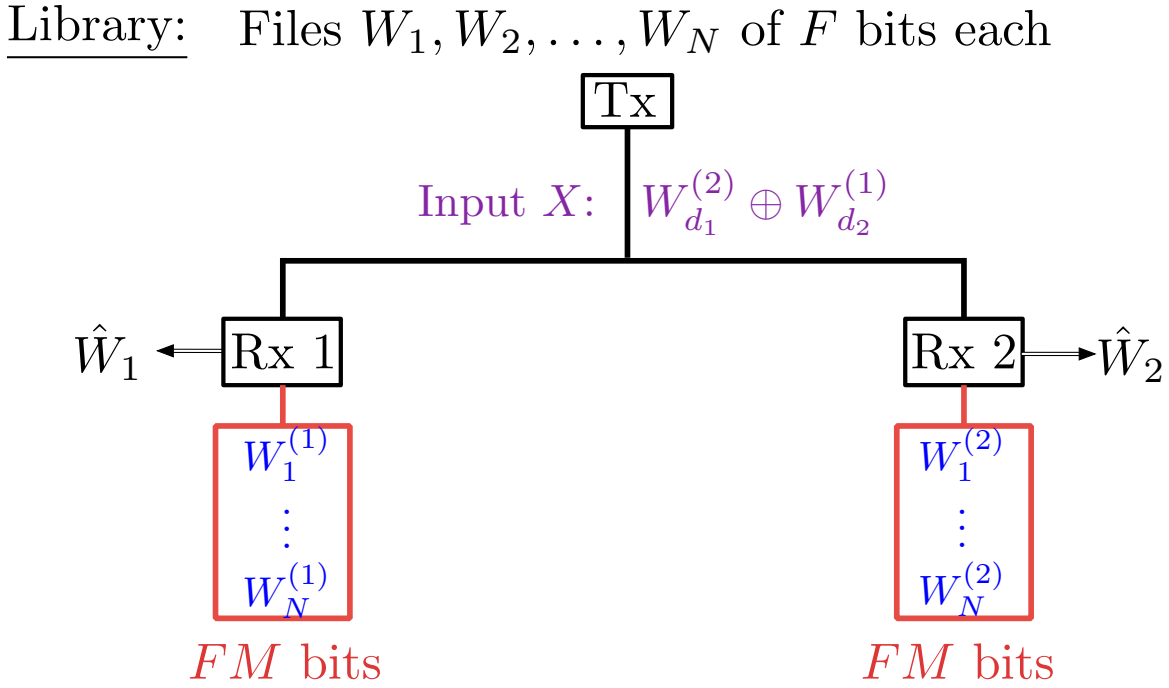
$$X = (W_{d_1}^{(2)},\ W_{d_2}^{(2)}, \dots,\ W_{d_K}^{(2)}).$$

  - If $K < N$, in the worst case:

$$X = (W_1^{(2)},\ W_2^{(2)}, \dots,\ W_N^{(2)}).$$

In the worst-case, rate needs to be $R = \min\{N, K\}\,\big(1-\frac{M}{N}\big)$.

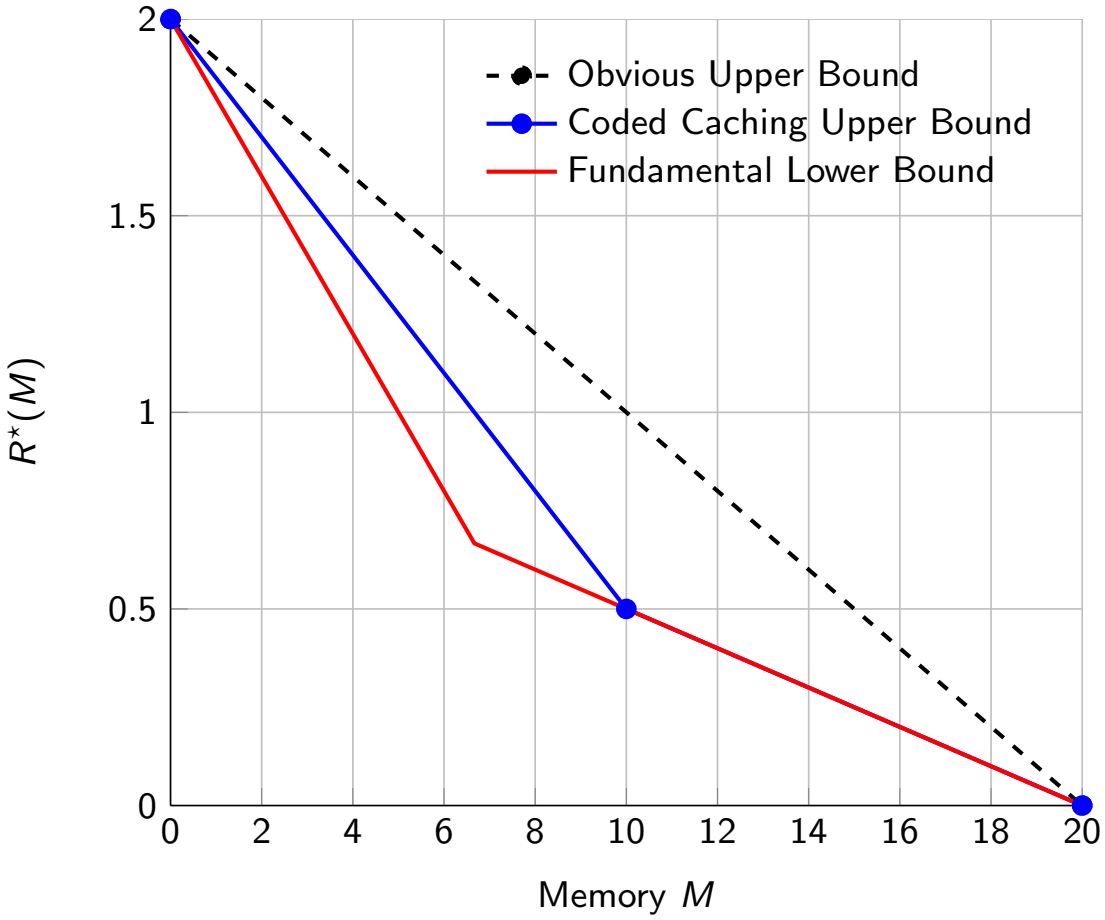
Coded caching for $K = 2$ Receivers [Maddah-Ali&Niesen 2014]



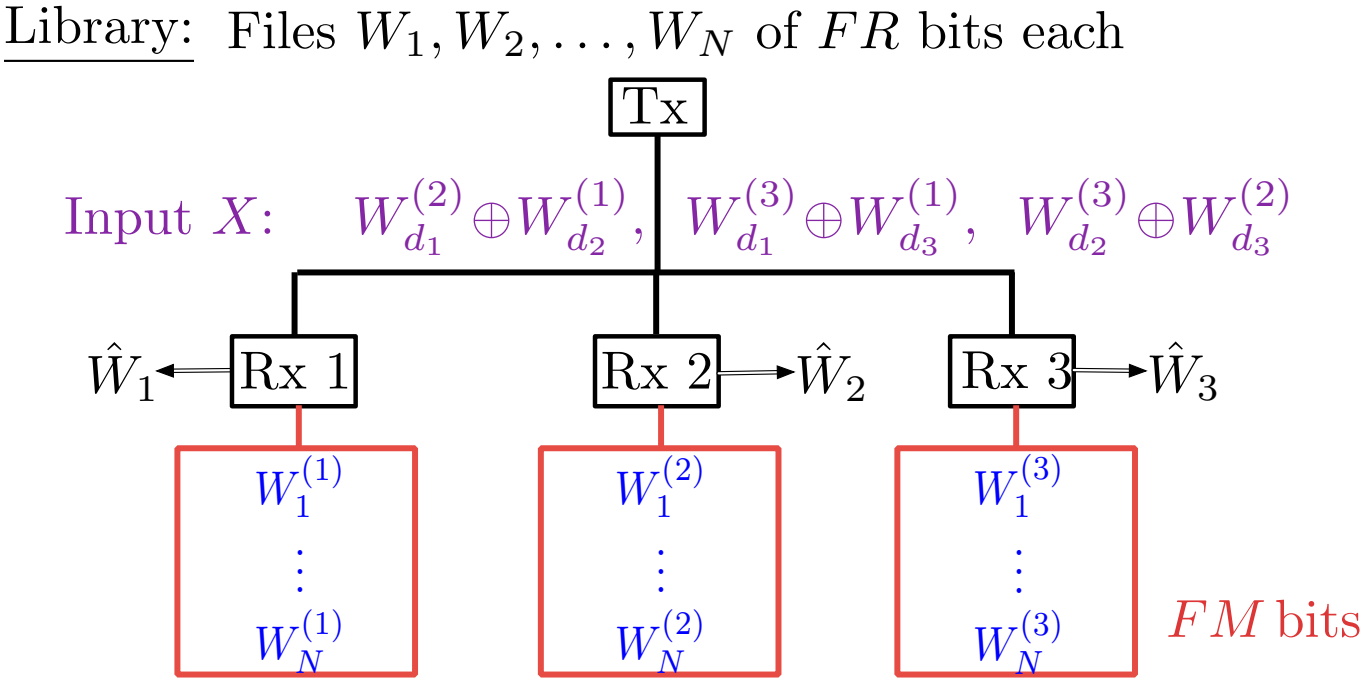
- Split W_d into two parts $(W_d^{(1)}, W_d^{(2)})$ each of $\frac{F}{2}$ bits
- For $d = 1, \ldots, N$: cache part $W_d^{(1)}$ at Rx1 and part $W_d^{(2)}$ at Rx2
- Delivery input $X = W_{d_1}^{(2)} \oplus W_{d_2}^{(1)}$

Achieves Rate-Memory Pair $\quad M = \frac{N}{2} \quad$ and $\quad R = \frac{1}{2}$.

Fundamental Limit and Bounds for 2 Users ($N = 20$ files)



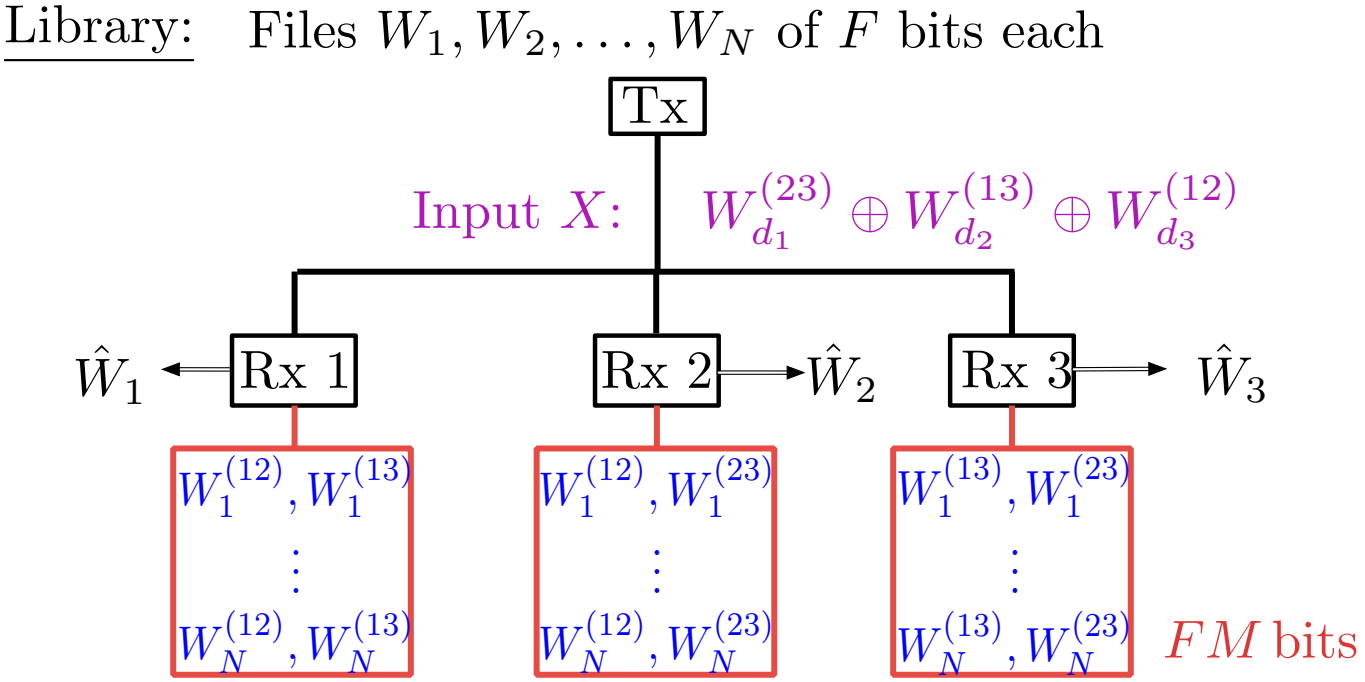
Coded caching for $K = 3$ Receivers, Parameter $t = 1$



- Split W_d into three parts $(W_d^{(1)}, W_d^{(2)}, W_d^{(3)})$ each $\frac{F}{3}$ bits
- For $d = 1, \ldots, N$: cache part $W_d^{(1)}$ at Rx1, part $W_d^{(2)}$ at Rx2, and part $W_d^{(3)}$ at Rx3
- Delivery input $X = W_{d_1}^{(2)} \oplus W_{d_2}^{(1)},\ W_{d_1}^{(3)} \oplus W_{d_3}^{(1)},\ W_{d_2}^{(3)} \oplus W_{d_3}^{(2)}$

Achieves Rate-Memory Pair $\quad M = \frac{N}{3} \quad$ and $\quad R = 1.$

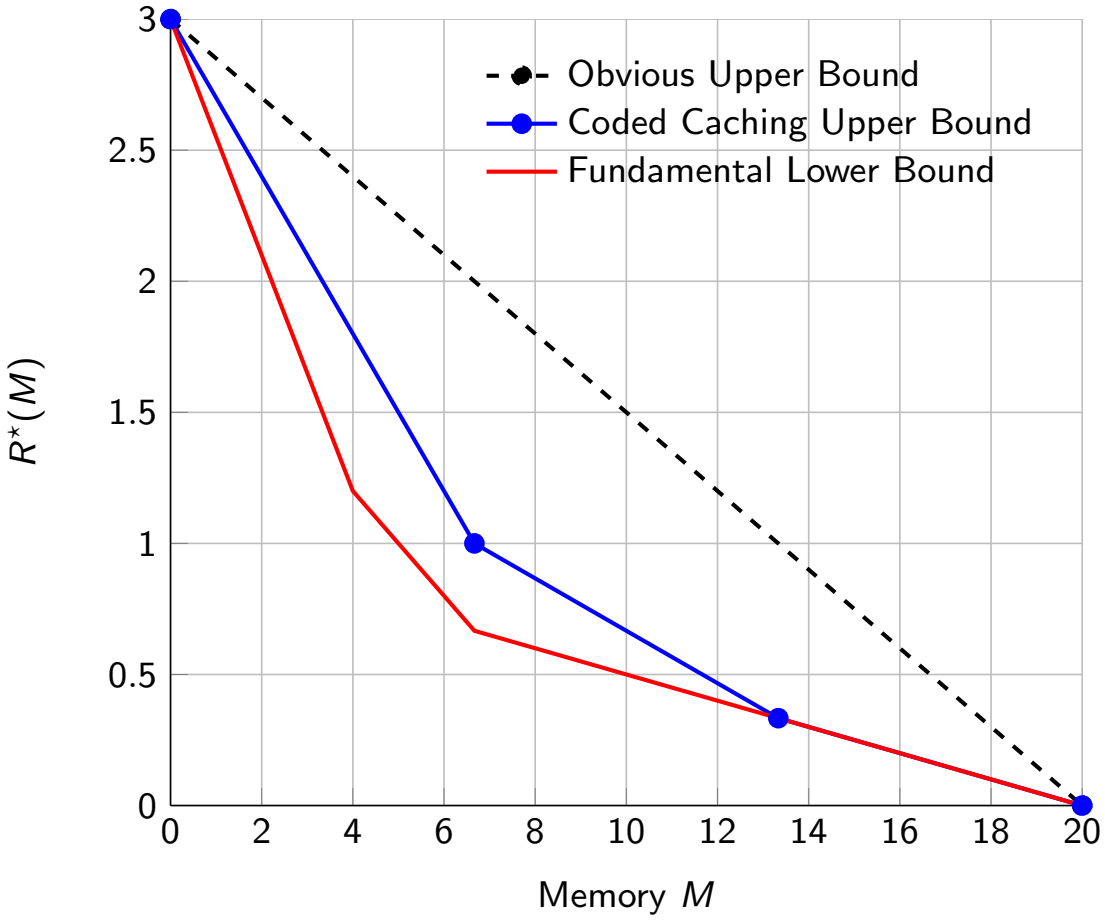
Coded caching for $K = 3$ Receivers, Parameter $t = 2$



- Split W_d into three parts $(W_d^{(12)}, W_d^{(13)}, W_d^{(23)})$ each of $\frac{F}{3}$ bits
- For $d = 1, \ldots, N$: cache part $W_d^{(12)}$ at Rxs 1 and 2, part $W_d^{(13)}$ at Rxs 1 and 3, and part $W_d^{(23)}$ at Rxs 2 and 3
- Delivery input $X = W_{d_1}^{(23)} \oplus W_{d_2}^{(13)} \oplus W_{d_3}^{(12)}$

Achieves Rate-Memory Pair $\quad M = \frac{2N}{3} \quad$ and $\quad R = \frac{1}{3}$.

Bounds for 3 Users ($N = 20$ files)



Coded Caching for K Users

- Parameter $t \in \{1, \dots, K-1\}$

- *Placement:* Split each W_d into $\binom{K}{t}$ parts and save each part at a different subset of receivers
  Let for each size-t subset $\mathcal{G}$ denote $W_d^{\mathcal{G}}$ the part of W_d placed in caches of all receivers in $\mathcal{G}$.

- *Delivery transmission:* For each set $\mathcal{S} = \{s_1, \dots, s_{t+1}\}$ of size $t+1$, send

$$W_{\text{XOR},\mathcal{S}} := \bigoplus_{\ell=1}^{t+1} W_{d_{s_\ell}}^{(\mathcal{S}\setminus\{s_\ell\})}$$

- *Delivery reception:* Receiver s_j has stored in its cache memory

$$W_{d_{s_\ell}}^{(\mathcal{S}\setminus\{s_\ell\})}, \qquad \forall \ell \in \{1, \dots, j-1, j+1, \dots, t\}.$$

  So, with $W_{\text{XOR},\mathcal{S}}$ it can recover $W_{d_{s_j}}^{(\mathcal{S}\setminus\{s_j\})}$.

  This way it can recover all missing parts of $W_{d_{s_j}}$.

Analysis of Coded Caching for K Users

- Fix parameter $t \in \{1, \dots, K-1\}$
- Each part of a file is of size

$$F \cdot \binom{K}{t}^{-1} \quad \text{bits.}$$

- Each receiver stores $\binom{K-1}{t-1}$ parts of each file. So the placement requires cache memory

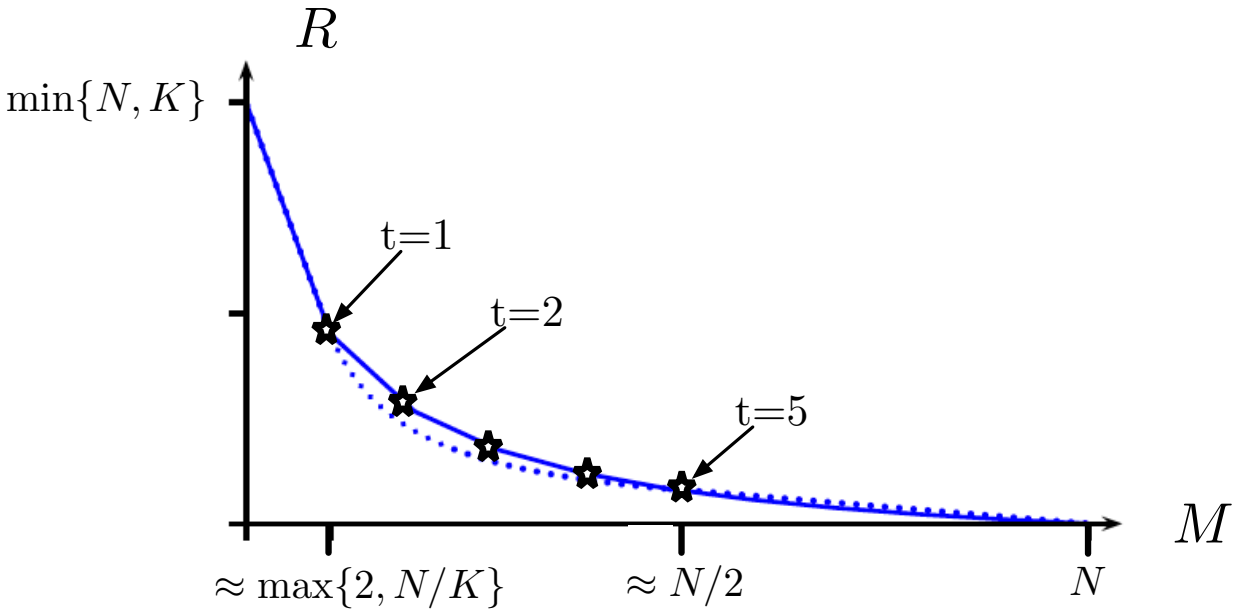
$$M = N \frac{\binom{K-1}{t-1}}{\binom{K}{t}} = N \frac{t}{K}. \qquad \text{(increasing in } t\text{)}$$

- Coded caching sends an XOR-message to each subset of $t+1$ receivers. So the total rate is

$$R = \frac{\binom{K}{t+1}}{\binom{K}{t}} = \frac{K-t}{t+1}. \qquad \text{(decreasing in } t\text{)}$$

Performance of Coded Caching

- $K = 6$



Coded Caching Upper Bound

For all $M \in \frac{K}{N} \cdot \{1, \ldots, K-1\}$:

$$R^\star(M) \leq K\left(1 - \frac{M}{N}\right)\left(1 + \frac{MK}{N}\right)^{-1}.$$

Bounds on $R^\star(M)$

From what we saw until now:

For all $M \in \frac{K}{N} \cdot \{0, 1, \ldots, K-1, K\}$:

$$R^\star(M) \leq K\Big(1 - \frac{M}{N}\Big)\Big(1 + \frac{MK}{N}\Big)^{-1}.$$

- For all M, lower-convex hull of these points forms an upper bound on $R^\star(M)$

Bounds on $R^\star(M)$

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- For all M, lower-convex hull of these points forms an upper bound on $R^\star(M)$

What we will prove in the following:

$$R^\star(M) \geq \max_{s \in \{1, \ldots, \min\{K,N\}\}} \left(s - \frac{s}{\lfloor N/s \rfloor} M\right).$$

Bounds on $R^\star(M)$

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What we will prove in the following:

$$R^\star(M) \geq \max_{s \in \{1, \ldots, \min\{K,N\}\}} \left(s - \frac{s}{\lfloor N/s \rfloor} M\right).$$

- Upper and lower bounds on $R^\star(M)$ match up to a factor of 5! (Proved numerically)

Proof of Lower Bound

- Consider only Receivers $1, \dots, s$, where $s \leq \min\{N, K\}$
- Consider demand vectors

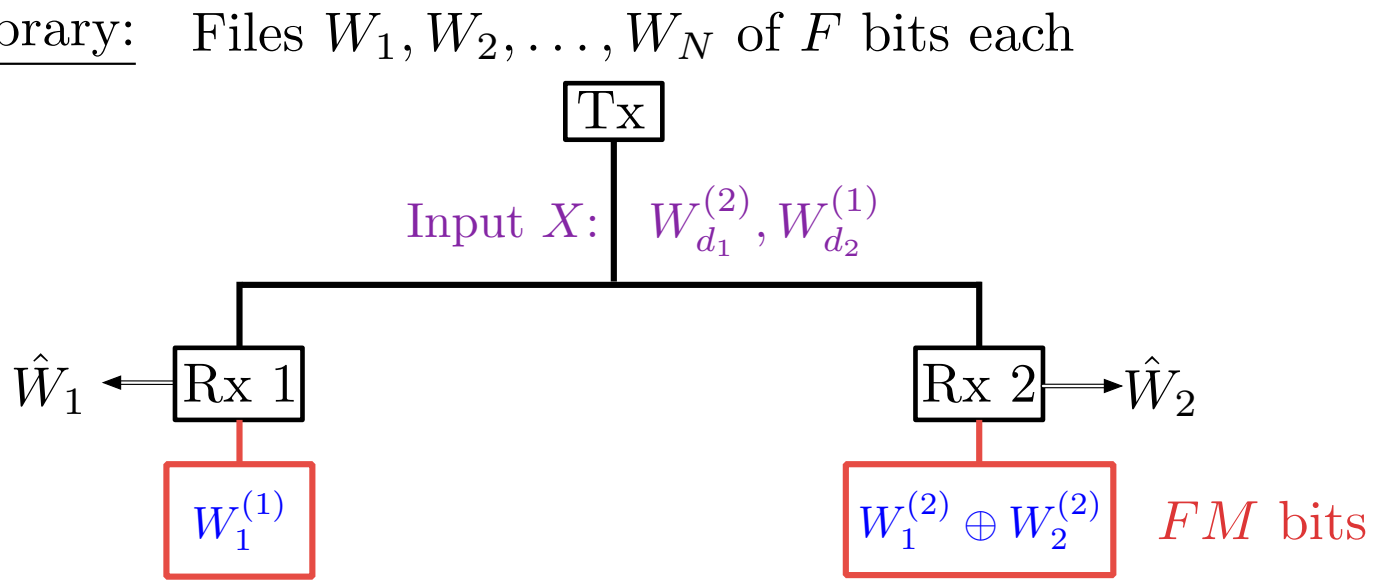
$$\begin{aligned}
\mathbf{d}^{(1)} &:= (1, \dots, s) \\
\mathbf{d}^{(2)} &:= (s+1, \dots, 2s) \\
&\dots \\
\mathbf{d}^{(\lfloor N/s \rfloor)} &:= ((\lfloor N/s \rfloor - 1)s + 1, \dots, \lfloor N/s \rfloor s)
\end{aligned}$$

- Let $X^{(\ell)}$ denote the delivery input for demand $\mathbf{d}^{(\ell)}$
- From $X^{(1)}, \dots, X^{(\lfloor N/s \rfloor)}$ and $Z_1, \dots, Z_s$ one can calculate $W_1, \dots, W_{\lfloor N/s \rfloor s}$:

$$\begin{aligned}
& H\big(X^{(1)}, \dots, X^{(\lfloor N/s \rfloor)}, Z_1, \dots, Z_s\big) \geq H(W_1, \dots, W_{\lfloor N/s \rfloor s}) \\
\iff \quad & H\big(X^{(1)}, \dots, X^{(\lfloor N/s \rfloor)}\big) + H\big(Z_1, \dots, Z_s\big) \geq H(W_1, \dots, W_{\lfloor N/s \rfloor s}) \\
\iff \quad & FR\lfloor N/s \rfloor + sFM \geq \lfloor N/s \rfloor sF \\
\iff \quad & R \geq s - \frac{s}{\lfloor N/s \rfloor} M.
\end{aligned}$$

Lower and Upper Bounds on $R^\star(M)$ can be Improved!

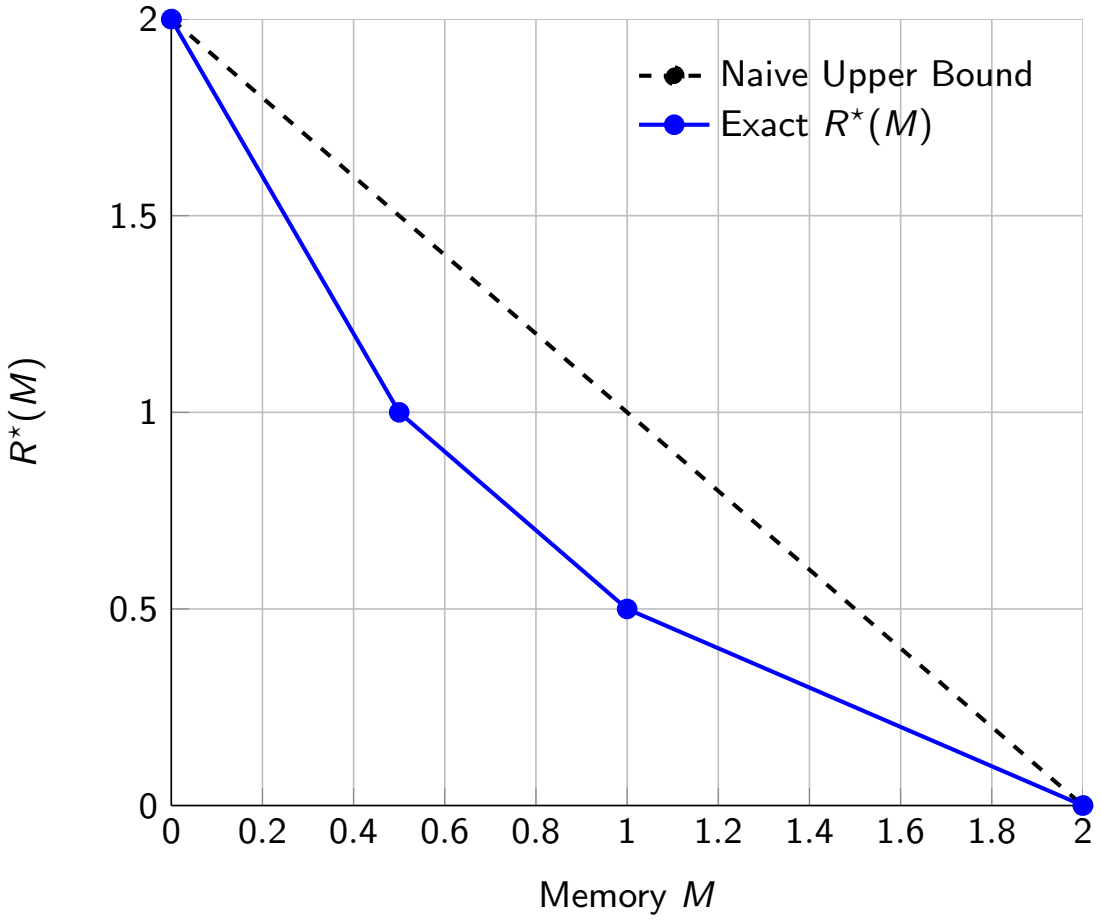
Example: $K = 2$ and $N = 2$:



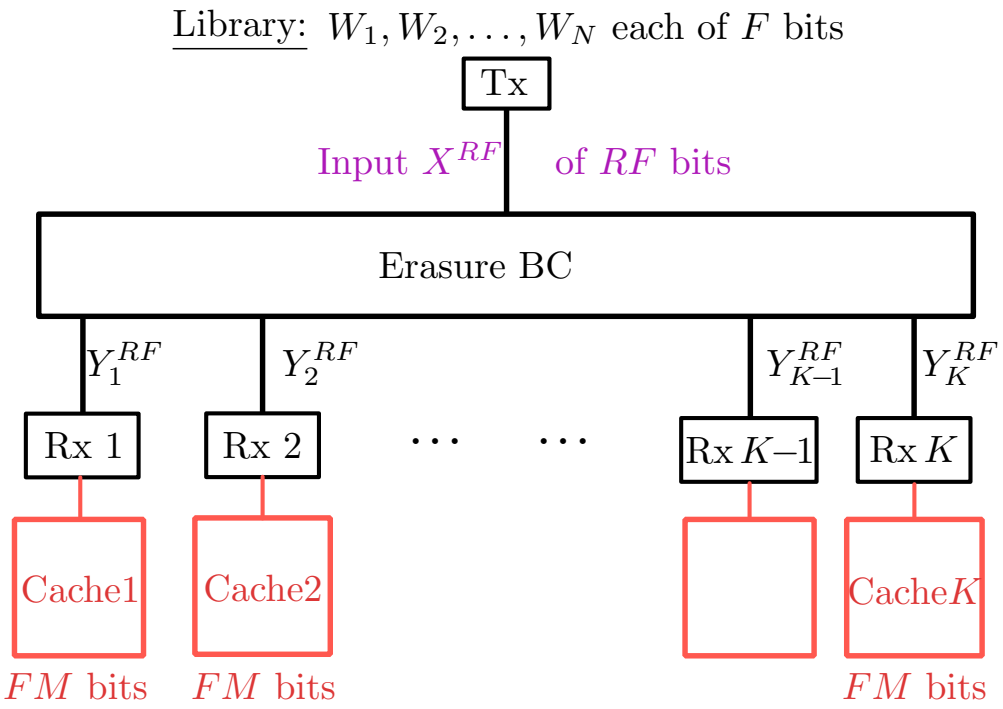
- Split W_d into two parts $(W_d^{(1)}, W_d^{(2)})$ each of $\frac{F}{2}$ bits
- For $d = 1, \ldots, N$: cache part $W_d^{(1)}$ at Rx1 and part $W_d^{(2)}$ at Rx2
- Delivery input $X = W_{d_1}^{(2)}, W_{d_2}^{(1)}$

Achieves Rate-Memory Pair $\quad M = \frac{1}{2} \quad$ and $\quad R = 1.$

Exact Rate-Memory Tradeoff $R^\star(M)$ for $K = N = 2$

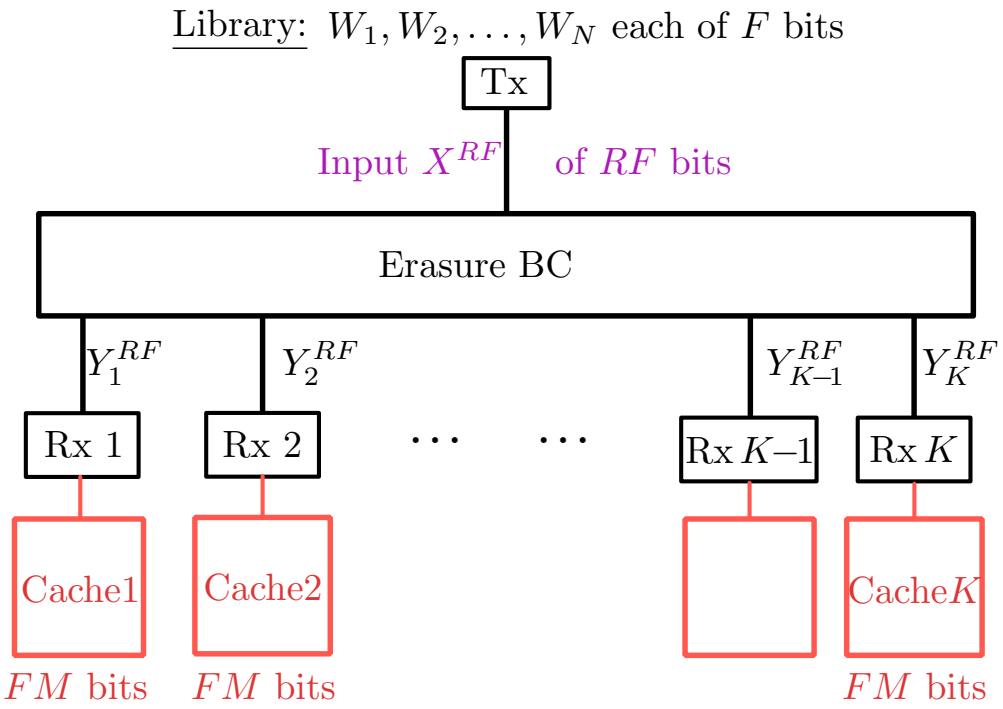


Delivery over Erasure Broadcast Channel (BC), Homework Exercise 1



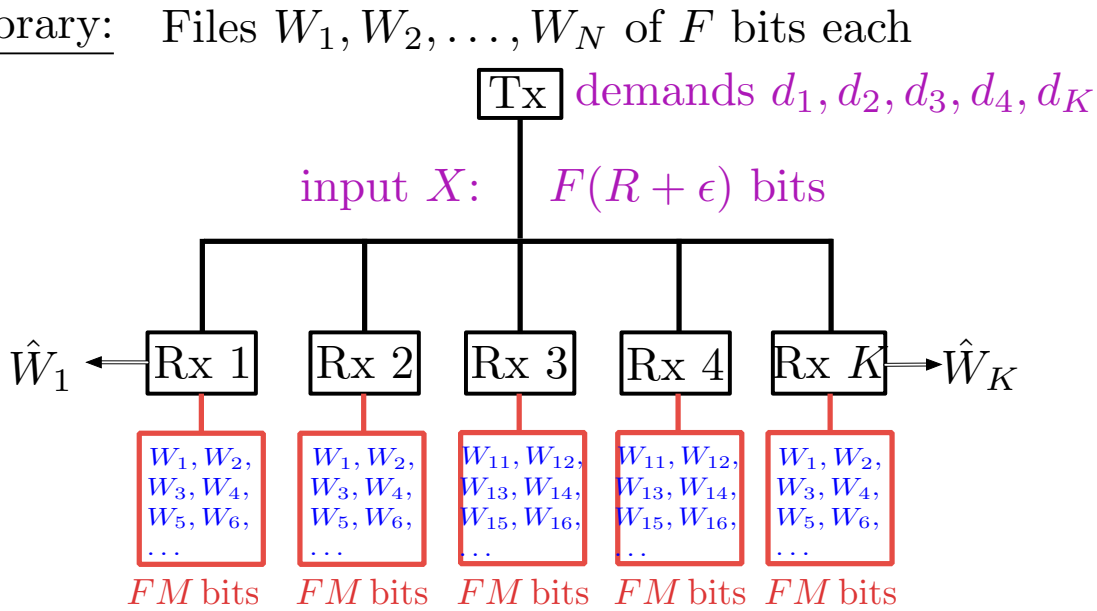
- Each transmitted bit erased at each receiver with probability $\delta > 0$
- Adapt the coded caching upper bound on $R^\star(M)$ to this setup!

Delivery over Erasure BC, Homework Exercise 2



- Erasure probability at receivers $1, \ldots, K_w$ is δ_1, where $K_w < K$
- Erasure probability at receivers $K_w + 1, \ldots, K$ is $\delta_2 < \delta_1$
- Adapt the coded caching upper bound on $R^\star(M)$ to this setup!

Decentralized Caching



- Cache placement $\quad Z_k = g(W_1, \ldots, W_N, \Theta_k)$,

  where Θ_k is a randomness known to everyone

- Delivery encoding $\quad X = f(W_1, \ldots, W_N, d_1, \ldots, d_K, \Theta_1, \ldots, \Theta_K)$

- Delivery decoding $\quad \hat{W}_k = \varphi_k(X, Z_k, d_1, \ldots, d_K, \Theta_1, \ldots, \Theta_K)$.

- Goal: $\quad \hat{W}_k = W_{d_k}$ for all $k = 1, \ldots, K$ $\quad$ with high probability

Fundamental Rate-Memory Tradeoff for Decentralized Caching

- (R, M) achievable if $\forall \epsilon > 0$ there exist $\{g, f, \varphi_1, \ldots, \varphi_K\}_{n=1}^{\infty}$ such that

$$\lim_{F \to \infty} \Pr\Big[\bigcup_{k=1,\ldots,K} \{\hat{W}_k \neq W_k\} \Big] = 0$$

- Fundamental Rate-Memory Tradeoff :

$$R^{\star}(M) = \min \big\{ R \colon (R, M) \text{achievable} \big\}$$

Decentralized Coded Caching Algorithm

- *Cache Placement:* Each receiver stores *randomly* $\frac{FM}{N}$ bits of each file W_d in its cache content

- Each Receiver tells everyone else *which bits of each file* it has cached. (But not the values of these bits.)

- *Delivery Encoding:*
  - Let the set $\mathcal{V}_{\mathcal{G}}$ indicate the bits (position and file) that are known to all receivers in $\mathcal{G} \subseteq K$ but not to any other receiver.

  - Send all demanded bits that are not cached anywhere

  - For $t = 1, \ldots, K-1$ use the coded caching delivery scheme of parameter t to send all demanded bits that are in sets

    $$\{\mathcal{V}_{\mathcal{G}} : |\mathcal{G}| = t\}$$

    Zero-padding might be required for this operation!

  - Delivery decoding similar to coded caching scheme, but again for many parameters t

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[2] M. A. Maddah-Ali, U. Niesen, "Decentralized coded caching attains order-optimal memory-rate tradeoff," *IEEE Trans. on Inf. Theory*.

Results for Decentralized Caching

Upper Bound $R^\star(M)$ for Decentralized Caching

$$R^\star(M) \leq K\left(1 - \frac{M}{N}\right) \min\left\{\frac{N}{KM}\left(1 - \left(1 - \frac{M}{N}\right)^K\right), \frac{N}{K}\right\}$$

(Proved on the board in class.)

- Above upper bound matches $R^\star(M)$ up to a factor of at lest 12. (Proved analytically in [2].)
- Above upper bound matches coded caching upper bound for centralized scenario up to a factor of 1.6. (Shown numerically.)